AQ8.3: Activity Questions 3 - Not Graded

This assignment will not be graded and is only for practice.
Level 1

1 point
Consider two vectors $a = (1, -2, 2)$, $b = (4, 0, -3)$a = (1, −2, 2), b = (4, 0, −3). Using the standard dot product as the inner product, the projection of aa in the direction of bb is given by

- $(\frac{-8}{25}, 0, \frac{6}{25})$(25−8, 0, 256)
- $(\frac{-8}{25}, \frac{6}{25})$(25−8, 256)
- $(\frac{8}{25}, 0, \frac{-6}{25})$(258, 0, 25−6)
- $(\frac{8}{25}, \frac{-6}{25})$(258, 25−6)

No, the answer is incorrect.
Score: 0
Accepted Answers:
$(\frac{-8}{25}, 0, \frac{6}{25})$(25−8, 0, 256)

1 point
Consider an orthonormal basis $\{\frac{1}{\sqrt{5}}(1, 2), \frac{1}{\sqrt{5}}(-2, 1)\}${ 5
√
1(1, 2), 5
√
1(−2, 1)} of R^2 R2. If $(3, 5) = c_1 v_1 + c_2 v_2$ (3, 5) = c1v1 + c2v2 where $c_1, c_2 \in R$c1, c2 ∈ R and $v_1 = \frac{1}{\sqrt{5}}(1, 2), v_2 = \frac{1}{\sqrt{5}}(-2, 1)$v1 = 5
√
1(1, 2), v2 = 5
√
1(−2, 1), then choose the correct option.

- $c_1 = \frac{13}{\sqrt{5}}, c_2 = \frac{1}{\sqrt{5}}$c1 = 5
√
13, c2 = 5
√
1
- $c_1 = \frac{-13}{\sqrt{5}}, c_2 = \frac{1}{\sqrt{5}}$c1 = 5
√
−13, c2 = 5
√
1
- $c_1 = \frac{13}{\sqrt{5}}, c_2 = \frac{-1}{\sqrt{5}}$c1 = 5

√
13, c2 = 5
√
−1

○
$c_1 = \frac{-13}{\sqrt{5}}, c_2 = \frac{-1}{\sqrt{5}}$c1 = 5
√
−13, c2 = 5
√
−1

No, the answer is incorrect.
Score: 0
Accepted Answers:
$c_1 = \frac{13}{\sqrt{5}}, c_2 = \frac{-1}{\sqrt{5}}$c1 = 5
√
13, c2 = 5
√
−1

1 point
Consider an orthonormal basis $\{(1,0,0),(0,1,0)\}$ {(1, 0, 0), (0, 1, 0)} for a subspace W W in R^3 R3. If $x = (1,2,3)$ x = (1, 2, 3) is a vector in R^3 R3, then which of the following represents a vector in W W whose distance from x x is the least? Consider dot product as the standard inner product.

○
$(2,4,0)$ (2, 4, 0)

○
$(3,4,0)$ (3, 4, 0)

○
$(4,5,0)$ (4, 5, 0)

○
$(1,2,0)$ (1, 2, 0)

No, the answer is incorrect.
Score: 0
Accepted Answers:
$(1,2,0)$ (1, 2, 0)

Level 2

Consider the inner product $\langle a, b\rangle = a_1 b_1 - a_1 b_2 - a_2 b_1 + 4 a_2 b_2$ ⟨a, b⟩ = a1b1 − a1b2 − a2b1 + 4a2 b2 where $a = (a_1, a_2), b = (b_1, b_2)$ a = (a1, a2), b = (b1, b2) are vectors in R^2 R2. Use this information to answer questions 4 and 5.

1 point
Let $x = (1,2)$ x = (1, 2). Find the projection of x x in the direction of $(3,4)$ (3, 4) using the inner product defined above.

○
$\frac{25}{49}(3,4)$ 4925(3, 4)

○
$\frac{11}{25}(3,4)$ 2511(3, 4)

○

$\frac{25}{49}(1,2)$4925(1, 2)

○

$\frac{11}{25}(1,2)$2511(1, 2)

No, the answer is incorrect.
Score: 0
Accepted Answers:

$\frac{25}{49}(3,4)$4925(3, 4)

1 point

Let $x = (1,2)$x = (1, 2), find the projection of xx in a direction perpendicular to $(3,4)$(3, 4).

○

$\frac{25}{49}(-4,3)$4925(−4, 3)

○

$\frac{11}{25}(4,-3)$2511(4, −3)

○

$(-\frac{26}{49}, -\frac{2}{49})$(−4926, −492)

○

$(\frac{26}{49}, -\frac{2}{49})$(4926, −492)

No, the answer is incorrect.
Score: 0
Accepted Answers:

$(-\frac{26}{49}, -\frac{2}{49})$(−4926, −492)

1 point

Consider an orthogonal basis $\{(1,2,1),(-2,0,2)\}${(1, 2, 1), (−2, 0, 2)} of a subspace WW, of the inner product space R^3 R3 with respect to the dot product. If $y = (1,2,3) \in R^3$ y = (1, 2, 3) ∈ R3, then find $Proj_W(y)$P rojW (y).

○

$(\frac{1}{3}, \frac{8}{3}, \frac{7}{3})$(31, 38, 37)

○

$(\frac{18}{\sqrt{6}} - \frac{32}{\sqrt{8}}, \frac{36}{\sqrt{8}}, \frac{18}{\sqrt{6}} + \frac{32}{\sqrt{8}})$(6

√

18 − 8

√

32, 8

√

36, 6

√

18 + 8

√

32)

○

$(\frac{1}{3}, -\frac{8}{3}, \frac{7}{3})$(31, −38, 37)

○

$(\frac{18}{\sqrt{6}} - \frac{32}{\sqrt{8}}, \frac{36}{\sqrt{8}}, -\frac{18}{\sqrt{6}} + \frac{32}{\sqrt{8}})$(6

√

18 − 8

√

32, 8

√

36, − 6

√

18 + 8

√

32)

No, the answer is incorrect.

Score: 0

Accepted Answers:

$(\frac{1}{3}, \frac{8}{3}, \frac{7}{3})(31, 38, 37)$

Check Answers

Your score is: 0/6